

Our Universe - Solar System and The Moon(Part - 1)

The universe is vast. The solar system forms a very small part of the universe. Our earth is one of the eight planets of the solar system. Yuri Gagarin was the first man to go into space. Valentina Tereshkova was the first woman to go into space. Rakesh Sharma was the first Indian man and later Kalpana Chawla became the first Indian woman to go into space.



Fig: Yuri Gagarin



Fig: Valentina Tereshkova



Fig: Rakesh Kumar



Fig: Kalpana Chawla

Moon

As the earth moves around the sun, it is accompanied by the moon. Even though the moon is earth's nearest neighbour in space, it is 3,84,400 km away from the earth's. The moon is 4500 million years old, the same age as the earth. The moon is called the earth's natural satellite because it revolves around the earth.

Why does the moon shine at night? The moon shines because it reflects the sun's light falling on it.

The surface of the moon is made up of plains, valleys and mountains. There are big holes called craters all over the moon. These craters are round and deep. A dark grey dust covers the surface of the moon.

Americans Neil Armstrong, Edwin Aldrin and Michael Collins were the first people to visit moon.



Fig: Neil Armstrong



Fig: Michael Collins



Fig: Edward Aldrin

They landed on the moon on 21 July 1969 in their spacecraft Apollo 11. Neil Armstrong was the first person to step on the moon.

The Moon and Tides

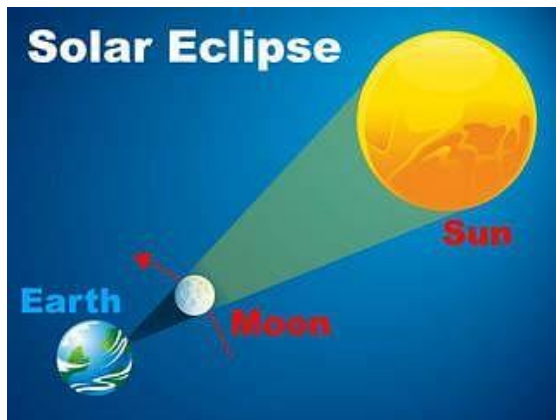
The rising and falling of the level of water in the sea are called tides. Tides are caused by the gravitational pulls of the moon and the sun. The moon attracts the ocean water with stronger force of attraction than that of the sun, because the moon is much closer to the earth than the sun.

Eclipse

An eclipse is a shadow that makes the sun or the moon totally or partially invisible for some time. The sun is the source of light and the earth and the moon are the objects that come in the path of this light and cast their own shadows. As the earth and the moon keep moving, their positions keep changing. Thus, their shadows also keep changing.

Solar Eclipse

Sometimes the moon comes between the sun and the earth. When this happens, the moon obstructs the light of the sun and casts its shadow on the earth. The people living in this part of the earth can see only a part of the sun or not see it at all. This is a solar eclipse. Once the moon moves out of this position, the sun can be seen again.



If the complete disc of the sun is covered by the shadow of the moon, it is called a total solar eclipse. If a part of the sun is hidden, it is called a partial solar eclipse.

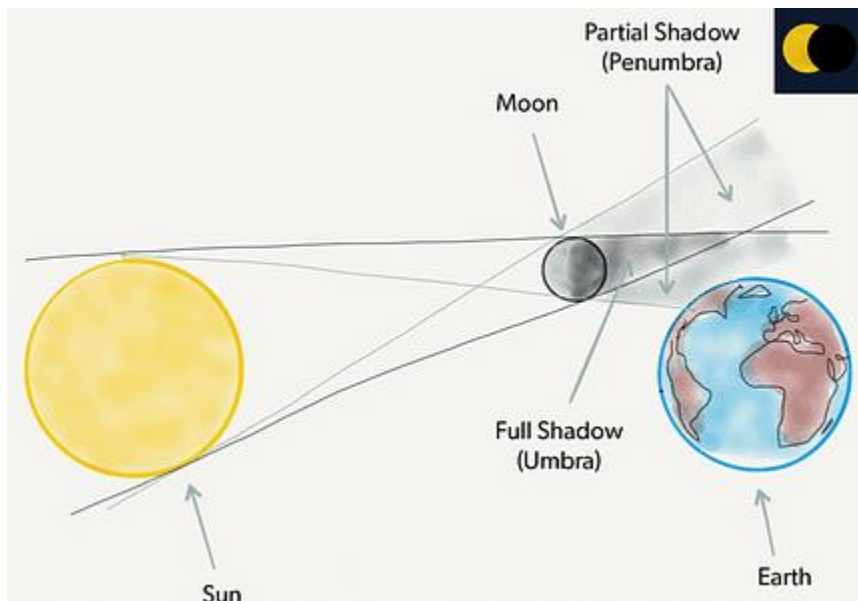
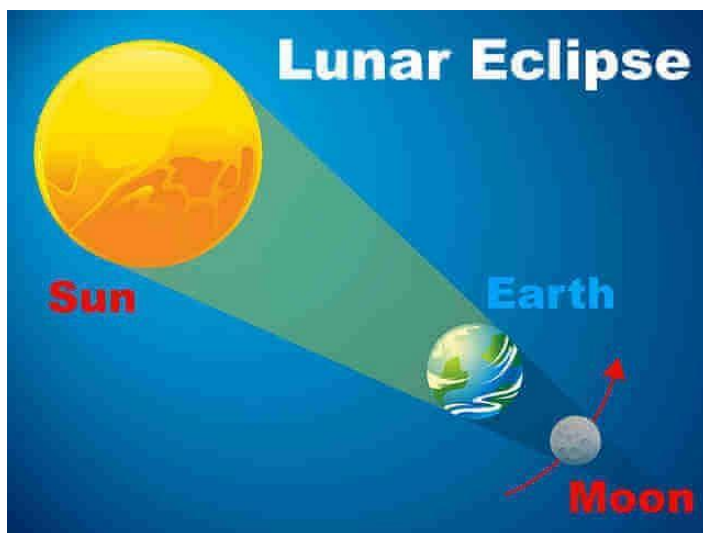


Fig: Partial Solar Eclipse

Lunar Eclipse

When the earth comes between the sun and the moon we cannot see the moon. This is because the earth comes in the path of the sun's light. You can see only a part of the moon or not see it at all. This is called a lunar eclipse. When the earth moves out of this position, you can see the moon again.



If the complete disc of the moon is covered by the shadow of the earth, it is known as a total lunar eclipse. If a part of the moon is hidden, then it is called a partial lunar eclipse.

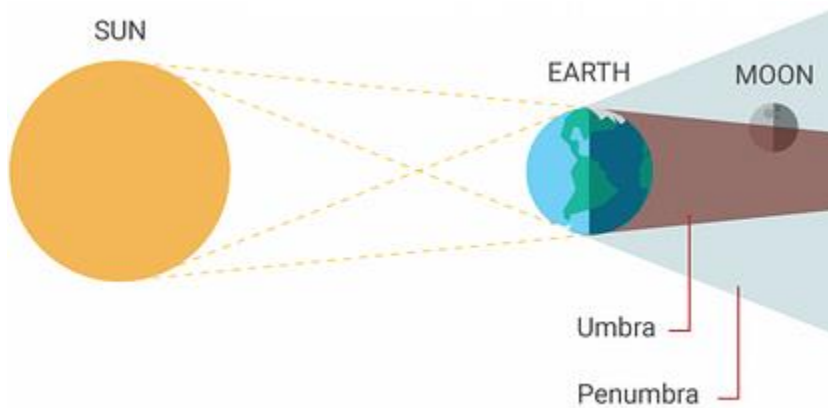


Fig: Partial lunar eclipse

Artificial Satellites

The moon is the earth's only natural satellite.

Many artificial satellites too revolve around the earth. These are man-made objects sent into space by scientists. Sputnik was the first man-made satellite. It was launched in 1957 by Russia. Since then, many satellites have been sent into space by different countries.



Fig: Artificial Satellite

Uses of artificial satellites

Weather satellites take pictures of cloud movements and help to forecast weather so that people can be warned about dangerous storms or cyclones before they actually happen. You are able to watch live cricket matches on television because of artificial satellites. Communication satellites are used for radio and TV transmission, telephonic communication, internet, etc. Artificial satellites can also give us different kinds of information about a particular area or region on earth. This includes forestry data, groundwater data, fishing areas in the sea, ocean studies, etc. Artificial

satellites also give us information about the space around the earth and other astronomical observations.

Some remarkable discoveries in deep space exploration in the recent years are as follows:

(1) India's first lunar probe, Chandrayaan-1, was the India's first major step in deep space exploration. It was launched by the Indian Space Research Organisation in October 2008, and operated till August 2009. It provided useful information about the features of the moon.

(2) India makes space history when the space agency of India, ISRO, successfully placed 'Mangalyaan' in Mars' orbit. With this India became the first country to successfully reach Mars orbit in the first attempt on September 24, 2014. It sent a lot of information about Mars.

(3) On November 12, 2014, 10 years after departing from earth, European Space Agency (ESA) lander Philae became the first probe to land on a comet. Philae left its spacecraft, Rosetta and landed on a comet named 67P/Churyumov-Gerasimenko. The contact from this space probe was lost on 15 November, 2014.